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Deprotonation of quinacridone on Ag(100) - an NIXSW/XPS study of the phase transition — •Lucas Hirschfeld[1], •Morris E. L. Mühlpointner[1], •Anna J. Kny[1], •Sergey Subach[2], and •Moritz Sokolowski[1] — [1]Clausius Institute for Physical and Theoretical Chemistry, University of Bonn, Germany — [2]Peter Grünberg Institut, Forschungszentrum Jülich GmbH, Germany

Quinacridone (QA) forms two different phases on the Ag(100) surface. When QA is deposited at room temperature, the formation of the α-phase with a point-on-line (*pol*) structure is observed. After annealing, the commensurate β-phase is irreversibly formed [1]. The mechanism, involving the loss of intermolecular hydrogen bonds, was not disclosed so far. Thus we have performed a detailed Normal Incidence X-ray Standing Waves (NIXSW)/XPS study.

The N1s XPS spectra exhibit a second peak shifted by 2.2 eV to lower binding energies when going from the α- to the β-phase. This reveals that the irreversible reaction from the α- to the β-phase is explained by a deprotonation of 50 % of the amine groups. The deprotonated N-atoms form stronger bonds to the Ag(100) surface, indicated by a drastic approach of the respective N adsorption height from 4.3 Å in the α-phase to 3.4 Å in the β-phase. These results show that the reaction is driven by the energy gain related to N/Ag interfacial bonds, which overcompensate the loss of intermolecular hydrogen bonds.

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